Alice Liang

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EDUCATION

Bachelors of Electrical Engineering (B.Eng), Year III | *McMaster University* — *Expected Apr. 2028*

- Pursuing a **Minor in Business and Concurrent Certificate in Sustainability**

WORK EXPERIENCE

SUSTAIN 1S03: Introduction to Sustainability Teaching Assistant — Jul. 2026 - Present
McMaster iLead Office — *Hamilton, ON*

- Educate the value and impact of sustainability initiatives, including **decarbonization and embodied carbon reports (LCA)**
- Foster interest in sustainability, responsible sourcing, and environmental, social, and governance topics to students by leading tutorials with over **90%** active participation
- Grade written **300+** assignments, attendance, and report feedback to students in a timely manner for **60+ students**, supporting ESG integration, report writing, and research

ENGINEER 1P13: Integrated Cornerstone Design Projects in Engineering Teaching Assistant — Jul. 2025 - Present
McMaster University Faculty of Engineering — *Hamilton, ON*

- Guide first-year engineering students on **Autodesk Inventor, Python, Q-arm,** and **3D printing** manufacturing
- Grade **50+** project deliverables and reports to provide feedback for students on **Word** and **Excel**
- Manage attendance of **100+ students** and oversee group engagement for students' design projects

Engineering Program Facilitator — Jun. 2026 - Aug. 2026
Shad Canada — *Lethbridge, AB*

- Designed, prepared, and delivered hands-on workshops for students teaching the engineering design processes
- Mentored Shad project groups of **20+ participants**, winning **Best Application of Theme** using an **Arduino prototype**
- Monitored and encouraged engagement and behaviours of participants, consulting with Program Directors for concerns

Software Development Blockchain Intern — Jan. 2026 - Apr. 2026
Community of Guardians — *London, ON*

- Building **blockchain-based** verification systems to align with United Nations Sustainable Development Goals (UNSDGs) by hashing to a public blockchain by deploying minimal smart contract (**Sepolia**), resulting in an increase of **70% user engagement**
- Programming with **Git** and **GitHub** to implement blockchain security with documentation

EXTRACURRICULAR EXPERIENCE

Vice President, Student Life (Prev. Director of Sustainability) — Mar. 2026 - Present
McMaster Engineering Society — *Hamilton, ON*

- Manage and recruited a portfolio of **10 directors/coordinators**, supporting a fast-paced yet team-oriented environment
- Organize and plan school-wide social events, facilitating inter-faculty relationships engaging outreach with other departments
- Develop and manage financial plans within **a $60,000+ budget**, including portfolio room booking and material costs

Vice President, Attendee Relations (Prev. Logistics) — Aug. 2024 - Present
DeltaHacks — *Hamilton, ON*

- Evaluate **1700+ hacker applications,** ensuring a diverse and qualified group of attendees
- Lead weekly team meetings to communicate objectives, resulting in a maintained project timeline
- Communicating with participants at help desks, resulting in streamlined check-ins and troubleshot technical issues

CAD Subteam Member — Sept. 2023 - Mar. 2024
First Robotics Canada Team 7902 — *Markham, ON*

- Engineered drivetrain and intake systems on **SolidWorks,** including chassis, bearing, and motors
- Evaluated **quality of materials** to be used on robots for manufacturing
- Communicated with team mentors and members to troubleshoot and **navigate changes** in game plan

PROJECTS

Bear With Me! devpost.com/software/bear-with-me — Nov. 2025
Hack Western 12 | *Hardware Developer*

- Operated on **Raspberry Pi 4**, encoding a plush with a responsive speaker and mic for children with speech development
- Compiled backend using **Python**, **Pygame, ElevenLabs, & Microsoft Azure API;** ensuring functions to be visible for SSH

The HocusFocus notion.site/Project-Three-Power-in-Community — Feb. 2024 - Apr. 2025
Integrated Cornerstone Design Projects in Engineering, McMaster University | *Project Manager*

- Modelled a magnifying tool to assist an individual with Usher's Syndrome, using **AutoCAD** and **Prusaslicer**
- Assessed for **quality control** during prototyping, coordinated fabrication schedules, and oversaw the successful assembly
- Presented as **Master of Ceremonies** for final showcase featuring **500+ guests** and using **Microsoft PowerPoint** and **Notion**

AWARDS & CERTIFICATIONS

McMaster Entrance Scholarship Recipient | *McMaster University* — Jun. 2024
Valedictorian | *Bur Oak Secondary School* — Jul. 2024
McGill University Shad Engineering Design Award for Best Pitch | *Shad Canada* — Jul. 2023

SKILLS

Languages: Python, C/ C++, HTML, CSS (Bootstrap, etc), Java, JavaScript, MATLAB, Verilog, LLMs
Hardware: PSPICE, Cadence, Raspberry Pi, Oscilloscopes, Multimeters, Breadboarding, 3D Printing (Prusa, BambuLab)
Other Software: SolidWorks, AutoCAD, Figma, SketchUP, Procreate, Photoshop, Microsoft Office Suite (Excel, Powerpoint, Word)